Sheth L.U.J. and Sir M.V. college
Functions in Python
Practical no.5

1) Python Function call:

```
def wish():
    print('Hello world! welcome')
wish()
print('outside the program')
```

```
Hello world! welcome
outside the program

=== Code Execution Successful ===
```

2) Python Function arguments:

```
def multi_numbers(num1,num2):
    multiplication=num1*num2
    print ("multiplication:",multiplication)
multi_numbers(5,6)
```

```
multiplication: 30

=== Code Execution Successful ===
```

3) Python variable scope:

i) Local variable:

```
def greet():
    #local variable
    message='Hello'
    print('local',message)
greet()
```

```
local Hello

=== Code Execution Successful ===
```

ii) Global variable:

```
#declare global variable
message='Hello'
def greet():
    print ('local',message)
greet()
print('Global',message)
```

```
local Hello
Global Hello

=== Code Execution Successful ===
```

iii) Nonlocal variable:

```
def outer():
    message='local'
    def inner():
        #declare the nonlocal variable
        nonlocal  message
        message='nonlocal'
        print ("inner:",message)
    inner ()
    print("outer:",message)
outer()
```

```
inner: nonlocal
outer: nonlocal

=== Code Execution Successful ===
```

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4) Python Global keyword:

```python
c=1
def add():
    #use of Global keyword
    global c
    c=c+2
    print(c)
add()
```

```
3

=== Code Execution Successful ===
```

5) Python Recursion:

```python
def factorial(x):
    #Recursive function
    if x==1:
        return 1
    else:
        return(x* factorial(x-1))
num=3
print("The factorial of",num, "is",factorial(num))
```

```
The factorial of 3 is 6

=== Code Execution Successful ===
```

6) Python modules:

```python
import math
print("the value of pi is ",math.pi)

```

```
the value of pi is  3.141592653589793

=== Code Execution Successful ===
```

7) Python main function:

```python
def main():
    print("Tasmiya Jahagirdar")
if __name__=="__main__":
    main()
```

```
Tasmiya Jahagirdar

=== Code Execution Successful ===
```

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